

Confidence Calibration for Low-Resolution Text Recognition Report

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Abstract—The reliability of the performance of text recognition under low-resolution is an important topic to studied. Knowing what kind of scene text condition, text recognition methods tend to assign relatively high confidence scores and bring an unreliable confidence estimate, can give us insight on when text recognition method’s confidence estimates become unreliable. In particular, the goal is to examine how visual challenges, for example blur image, low contrast, complex ground. affect both recognition accuracy and confidence calibration. This study focuses on evaluating whether existing recognition models know that they are uncertain and whether lightweight calibration strategies can improve reliability under degraded visual conditions. The project files can be found at: <https://github.com/TheRelaxingBeginning/Confidence-Calibration-for-Low-Resolution-Text-Recognition>

I. INTRODUCTION

A. Background

Text recognition is an important technology that improves our ability to process text; for example, we can copy and paste a web link straight from an image. Over the years, many text recognition methods have achieved strong performance in standard text recognition, for example in extracting words from a very clear text. However, recognizing text in a degraded image is still tough, and getting a degraded image is usually inevitable in real world applications. For example, text is often captured under harsh conditions, such as blur, orientation, shape, low resolution, etc. [1]. These factors can significantly reduce the accuracy of recognition. Many previous text recognition research has focused on improving recognition accuracy, but accuracy alone cannot fully describe model reliability. It is also important to know whether the confident score provided by the model is indeed trustworthy. The study of confidence calibration can help us understand whether the predicted confidence score of a model reflects the true probability of correctness. Therefore, this project is a comparison study for existing reliability models and also aims to examine whether simple calibration strategies can reduce overconfident errors without reducing recognition performance.

B. Research Question

In this study, we focus on three main research questions. First, we test whether the recognizers generally have high confidence, as they are producing overconfident errors at low resolution (incorrect predictions with a high confidence score). Secondly, it aims to find which types of image degradation or visual condition (blur, low contrast, or complex background) are more likely to produce these high confidence errors. Third, without modifying the recognizers structure, is a lightweight calibration possible to improve

its confidence reliability . Overall, these questions focus on a better understanding of how accurately the recognizer performs under low resolution conditions and how accurate their confidence is.

II. METHOD

A. Dataset Information

In this study, we used the TextZoom dataset [1], which contains paired low-resolution and high-resolution scene text images. In addition, the data set contains labels and the train test split for the text images. Thus, it can be considered for both recognition evaluation and confidence analysis. In addition, the test set is split into three difficulty classes: easy, medium, and hard. The test/medium subset is primarily used, representing a balance between easy and hard. And for calibration purposes, the train2 training set is used applied to map from raw confidence to calibrated confidence.

B. Recognizer

In this study, three recognizers are being evaluated, namely CRNN, ASTER, and MORAN. These three recognizers differ from each other in terms of their complexity and method of treatment for irregular scene text. Hence, they offer an appropriate basis to evaluate both the recognition accuracy and confidence reliability.

1) *CRNN*: CRNN is used as a baseline model for scene text recognition in this study. It is a combination of a convolutional neural network (CNN) for visual feature extraction, a recurrent neural network (RNN) for sequential modeling of text output, and a Connectionist Temporal Classification (CTC) decoder for final prediction [2]. In terms of methodology, CRNN does not produce a character at a time. Instead, it produces a sequence of character probabilities over time, with blank removal and merging of repeated characters into a single word by the CTC decoder.

Due to its relatively simple structure, CRNN is efficient and a well-regarded benchmark, though it is sometimes less robust against input text that is distorted or very irregular.

2) *ASTER*: ASTER is short for Attentional Scene Text Recognizer with Flexible Rectification, is a stronger recognizer designed for more challenging scene text images. It is a combination of a rectification network to correct irregular text and an attention-based recognition network to predict the text sequence [3]. In terms of methodology, the rectification network tries to correct irregular text shapes before recognition. And after this step, the attention-based decoder will predict the text sequence character by character. And since it has a rectification step, it makes ASTER better

suiting for slanted, distorted text, which is common in real-world images. Thus, compared to CRNN, ASTER has a better architecture for irregular scene text recognition and a good comparison to analyze recognition performance and confidence reliability.

3) *MORAN*: MORAN is short for Multi Object Rectified Attention Network, it also has a rectification step and an recognition network for predicting text sequence [4]. In terms of the methodology, just like ASTER, it first tries to normalize the distorted text with a multi-object rectification network and then predicts the output sequence step by step with attention-based sequence recognition network. As it is designed for visually complex and distorted scene text, it is a stronger model for comparison in low-resolution settings compared to CRNN. Since it has similar design ideas with ASTER but differs in architecture, it provides an additional perspective on how recognizer design may affect both recognition performance and confidence reliability.

C. Confidence Definition

To compare CRNN, ASTER and MORAN based on their performance, it is necessary that the three models can generate prediction output in a similar form. That would allow them to be compared. Due to the fact that they have different decoders and use different alphabets, I had to normalize the prediction results into a common format so that they could be compared. First, after making the native predictions from each model (i.e. reducing to lowercase and removing all non-alphanumeric characters) only "a-z" and "0-9" remained. So, the predictions from each model were evaluated within the same character space and thus were subject to the same measure of accuracy. We also created a unified word-level confidence metric. In order to calculate this metric for each of the recognizers, we evaluate the recognized character sequence one token at a time. At each position, we take the maximum probability of the token as the confidence of the selected token. Then we apply the geometric mean to the word-level confidence metrics to create a single confidence value for the entire recognized word sequence, excluding tokens that do not belong to the shared alphabet. This is because applying a numerical mean to all tokens would cause a single token with low confidence to significantly decrease the overall confidence of the entire word.

D. Evaluation Metrics

In this study four metrics are used, namely: Word Accuracy, Expected Calibration Error (ECE), Brier Score, and High-Confidence Error Rate (HCER). These metrics are designed to capture different aspects of the model behavior. Word Accuracy reflects shows whether a recognizer predicts the text correctly, and the other three metrics focus on whether the confidence scores of the model are trustworthy.

1) *Word Accuracy*: The accuracy of words represents the proportion of words that were predicted by the recognizer that matched the truth label. It can reflect the model's ability to correctly predict text in a straightforward and simple way. However, relying solely on word accuracy does not

$$ECE = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|$$

Fig. 1: ECE Formula

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2$$

Fig. 2: Brier Score Formula

provide a full view of how good or bad a model is. A model could still produce a relatively high word accuracy, yet consistently generate high confidence when predicting incorrectly. Therefore, word accuracy is required but is insufficient to understand the total quality of a recognizer.

2) *ECE*: The Expected Calibration Error (ECE) measures the difference between predicted confidence and empirical accuracy across different confidence bins. An example would be that if a model assigns confidence scores of around 0.9, then the model should have an ideal accuracy rate of approximately 90%. If there is a significant discrepancy from this value, either higher or lower, the model is either overconfident or underconfident. Lower ECE values indicate that the model is well-calibrated, thus confidence estimates can be representative of reality. ECE is critical in this research study since we want to evaluate whether the recognizer is accurate, but also whether we can rely upon the confidence estimates.

3) *Brier Score*: The Brier score measures the overall squared error between the predicted confidence and the true correctness label. In this case, the correctness label is represented as 1 when the prediction is correct and 0 when the prediction is incorrect. Thus, Brier Scores assess confidence at an individual sample level rather than to assessing the relationship between confidence and accuracy at a bin level. Samples that are incorrectly predicted with high confidence are heavily penalized, whereas samples where predictions are correct and were estimated with confidence close to 1 receive less penalty. Consequently, lower Brier scores are a measure of how good your model was at predicting the reliability of those probabilities.

4) *HCER*: The high confidence error rate (HCER) measures the proportion of incorrect predictions that are still assigned high confidence (greater than 0.8). The purpose of this metric is particularly relevant to this study because it directly captures the type of error that is most concerning in practical applications: The model generates an incorrect output and produces an extremely high level of confidence. Thus, such errors can be more harmful than low confidence mistakes. Compared to ECE and the Brier Score, HCER is focused on overconfident but wrong prediction instances and serves as an explicit indication of overconfidence.

$$\text{HCER} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}(p_i \geq 0.8 \wedge y_i = 0)$$

Fig. 3: HCER Formula

E. Calibration Method

Platt scaling is a logistic calibration method. It is used in this study as a lightweight post processing method to improve confidence reliability. A primary benefit of using Platt Scaling for improving confidence credibility is its ability to be easily added after to existing recognizers, as it is only applied after prediction. Therefore, no modification to the model architecture is required. The confidence scores produced by the recognizer remain unchanged; however, the confidence score associated with each prediction will now have been adjusted by applying the calibration model. Therefore, recognition accuracy should not be affected; whereas calibration metrics, such as ECE, Brier Score, and HCER, may be improved. Thus, logistic calibration provides an efficient means to increase trustworthiness without affecting the underlying recognizer.

The calibration model was trained on the train2 split of TextZoom dataset and then applied to both the test/medium set. More specifically, raw confidence scores are first extracted from the recognizer output in train2, and these scores are paired with correctness labels to fit the logistic calibration model. After training, the learned mapping was utilized to map the raw confidence values obtained from the test/medium set to calibrated confidence scores. This allows the study to evaluate whether calibration improves the reliability of confidence estimates on unseen data.

III. OVERALL RESULTS

A. Recognition Performance

As shown in Figure 5, Among the three models, ASTER achieves the highest word accuracy (0.4111), MORAN is second (0.3664) and CRNN is worst (0.2055). This result shows a clear performance difference between the recognizers. Overall, there is an obvious performance gap for the recognizers. In addition, it seems that the models using rectification and strong sequential modeling are generally more successful when dealing with low resolution images containing text.

B. Confidence Reliability Comparison

As shown in Figure 6,7,8. These three models also show clear differences in confidence reliability. ASTER achieves the best overall calibration performance, with the lowest ECE, the lowest Brier Score, and the lowest High-Confidence Error Rate. MORAN falls somewhere in the middle on all three measures; while CRNN showed a very low level of reliability as well as the highest error rate from over-confidence. Thus, in this study, it appears that there is a strong correlation between recognition performance and the reliability of confidence ratings.

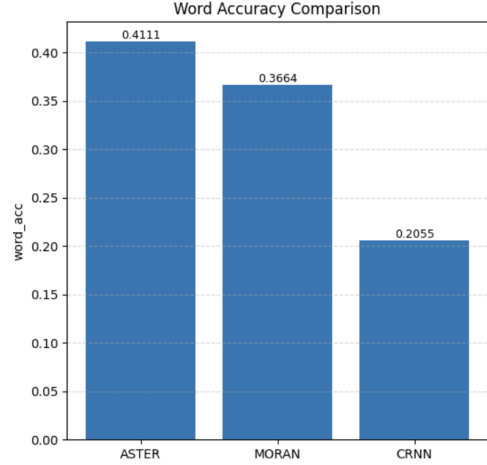


Fig. 4: Word Accuracy Result

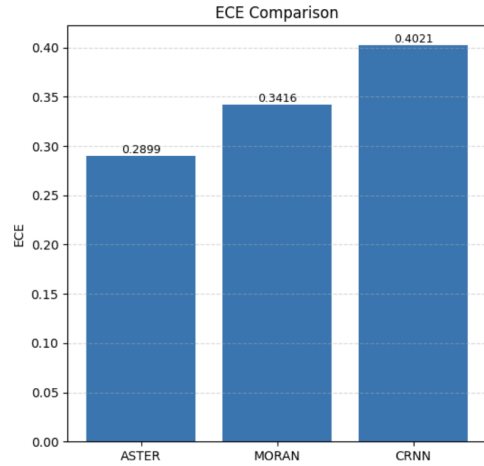


Fig. 5: ECE Result

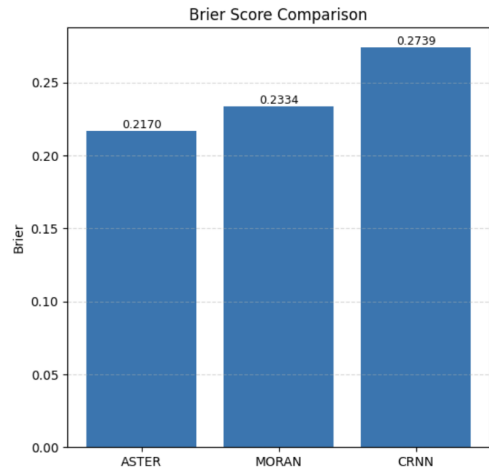


Fig. 6: Brier Score Result

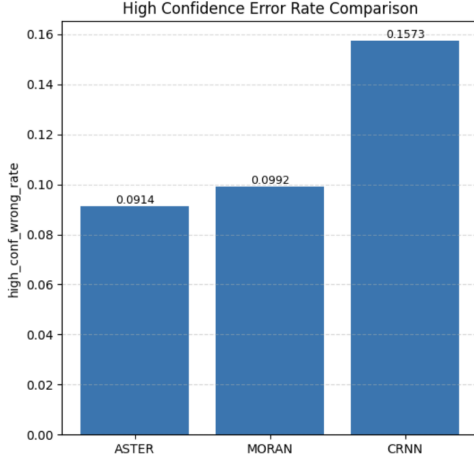


Fig. 7: HCER Result

IV. GROUP ANALYSIS

In order to examine the relationship between image degradation type and overconfident errors, three study groups are classified, namely: Blur, Low Contrast, Complex Background group. The corresponding scores are used to assign samples to each group. For Blur and Low Contrast, samples with scores below the 25th percentile of the score distribution are classified into the corresponding group. For Complex Background, samples with scores above the 75th percentile are classified into the complex background group. HCER is used as the metric since it is an explicit indication of overconfidence.

A. Blur

This study group is classified by blur score, which is measured using Laplacian variance. The intuition is that clear images typically display sharper edges with greater local changes in intensity than blurred images. Therefore, clear images should provide larger Laplacian responses. Blurred images typically will be less distinct and therefore smooth out both edges, so their Laplacian responses are generally smaller and less variable. Therefore, a lower Laplacian variance is used as an indicator of a stronger blur. As shown in Figure 9, the overall trend is not enough to support a strong conclusion. Although the non-blur group shows higher HCER values than the blur group in this result, the blur group is very small, which may limit the strength of the evidence. Therefore, due to these limitations, we cannot definitively state that blur has an impact on high-confidence errors based on the findings of this single analysis.

B. Low Contrast

This study group is classified by low contrast score, which is measured using the grayscale standard deviation. The intuition is that when the contrast between text and background is low, the variation in pixel intensity is smaller, making the text less visually distinct from its background. Therefore, a lower grayscale standard deviation can be used as an indicator of a stronger low-contrast degradation. As shown in Figure 10, all

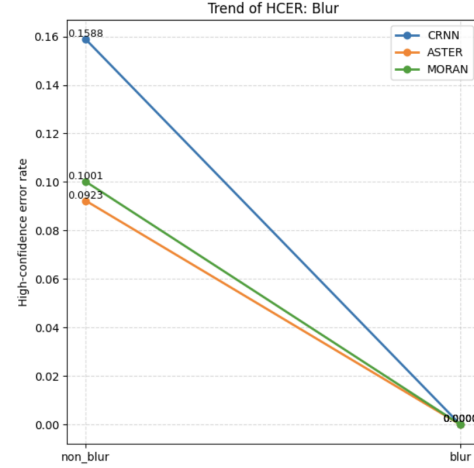


Fig. 8: Blur Group Comparison

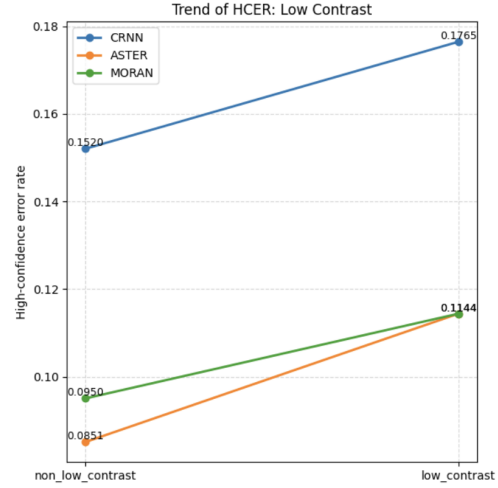


Fig. 9: Low Contrast Group Comparison

three models show higher HCER values in the low contrast group than in the non-low-contrast group. This suggests that low contrast can increase the chances of overconfident errors. The most likely reason for this is that when there is little or no brightness difference (low contrast) in the text and the surrounding area, the recognizer will receive less clear information about what it is seeing, but it will probably report a higher confidence level about its incorrect prediction. So, based on these findings, low contrast has a more consistently strong influence on predictions than blur.

C. Complex Background

This study group is classified by complex background score, which is measured using a combination of average edge density and image entropy. Edge Density is measured from the canny edge detection map as the ratio of edge pixels to total number of pixels within an image; Entropy is calculated from a grayscale image's histogram using shannon entropy and then scaled by dividing the result by 8.0. Images with much visual structure (i.e., cluttered patterns) and/or highly irregular backgrounds will have relatively high edge

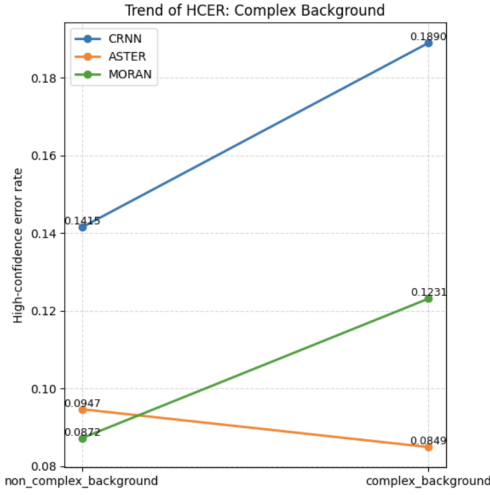


Fig. 10: Complex Background Comparison

density scores and relative high entropy scores. Therefore, higher complex background scores are used to indicate higher levels of background complexity. Figure 11 indicates that both CRNN and MORAN have significantly larger HCER values in the complex background group compared to the non-complex background group. ASTER does not show the same type of relationship and actually shows a slight decrease in the HCER value when moving into the complex background condition.

V. CALIBRATION RESULTS

This part was aimed to assess the effectiveness of the calibration method by evaluating the differences in how well the models performed both before and after the application of logistic calibration. It is imagined that Calibration does not change the predicted text, but rather to make the confidence scores better reflect the true probability of correctness. Because the calibration is a post processing step, no significant impact upon recognition accuracy should be found. As shown in Figures 12–14, logistic calibration leads to clear improvements in confidence reliability for all three recognizers. The ECE decreases substantially after calibration, indicating that the predicted confidence becomes more consistent with the empirical accuracy. The Brier score also decreases, showing that the overall confidence quality as a probability estimate improves. In addition, HCER decreases, which means that the number of incorrect predictions assigned high confidence becomes smaller. In general, these results indicate that a lightweight calibration process could increase trustworthiness. Therefore, logistic calibration appears to represent a scalable solution for increasing confidence reliability for low resolution scene text recognition in later work.

VI. CONCLUSION

The results of this research demonstrate that there are indeed issues with overconfidence in low resolution image-based character recognition. Overall, ASTER performed better than the other two recognizers (MORAN and CRNN)

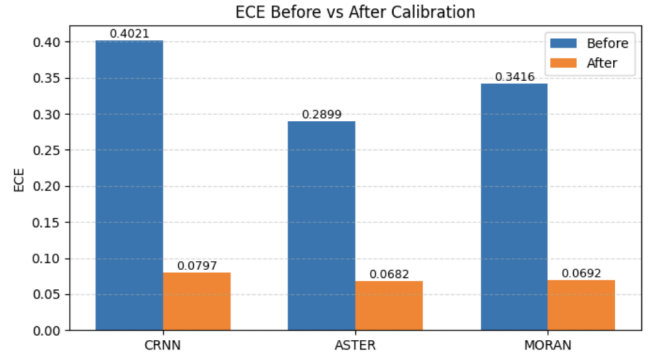


Fig. 11: ECE Before & After

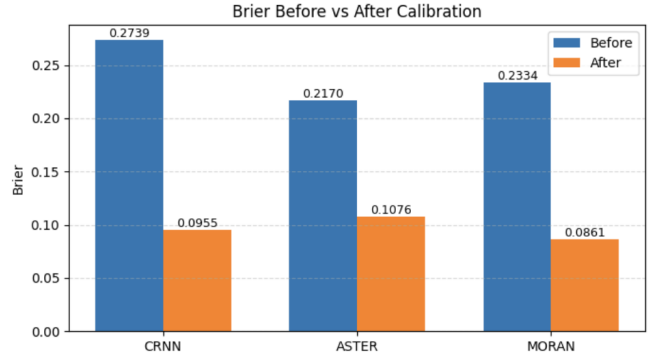


Fig. 12: Brier Score Before & After

in both average word accuracy and confidence reliability. MORAN was in the middle; CRNN had the lowest scores on both metrics. Low contrast is shown to be the most reliable and consistent risk factor for HCER across all recognizer models. Although complex backgrounds affect confidence reliability, their effects differ depending on which recognizer model is being used, as ASTER appears to be significantly more resistant to the negative effect of complex backgrounds compared to CRNN and MORAN. Then the results of the calibration indicate that the confidence reliability can be improved by logistic calibration. Overall, these results support the conclusion that there exists an interplay between how models are built and how they are calibrated after

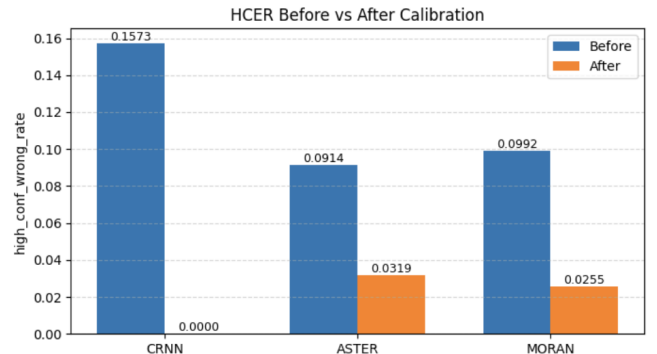


Fig. 13: HCER Before & After

training to create more trustworthy low resolution scene text recognition systems.

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